

PCA

Principal Component Analysis.

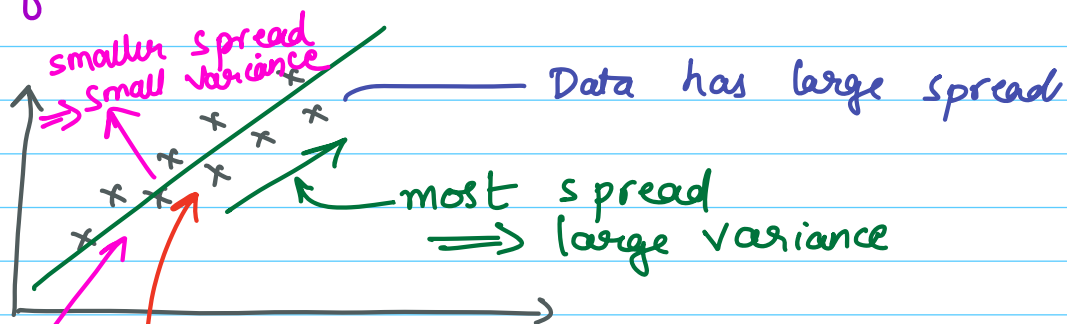
Most important concepts in ML.

Dimensionality extremely high! $\rightarrow \vec{x}_i$ $\rightarrow$ very high dimensional
 $\rightarrow$ facial images 256×256 pixels
 $\rightarrow$ vector of size $256^2 \times 1$

High dimensionality $\Rightarrow$ Analysis is Difficult!

$\rightarrow$ Reduce Dimension
 $\Rightarrow$ 'Dimensionality Reduction'

$\rightarrow$ Extract relevant features
 $\Rightarrow$ feature extraction.



Data can be projected along this direction.

How to determine the directions w/ largest Variance?

PCA gives the principal directions
 $\rightarrow$ project data along principal directions
'principal components'.

These are uncorrelated or orthogonal to each other.

PCA Procedure

consider data set

$\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n$ $m \times 1$
 n points
→ Each vector of size m

$$\bar{x}_i = \tilde{x}_i - \bar{\mu} \quad \text{sample mean}$$

$$\bar{\mu} = \frac{1}{n} \sum_{i=1}^n x_i$$

sample mean

Covariance Estimate :

$$R = \frac{1}{n-1} \sum_{i=1}^n \bar{x}_i \bar{x}_i^T$$

P.S.D matrix

$$\lambda_i \geq 0$$

Eigenvectors for distinct λ_i 's are orthogonal.

Principal Direction can be found as follows:

Let $\bar{v}_1$ be the direction of the largest principal component.

$\Rightarrow \bar{v}_1$ is the direction corresponding to the largest variance or maximum spread.

$\bar{v}_1^T \bar{x}_i$ — projection of $\bar{x}_i$ along $\bar{v}_1$.

$$\text{variance} = \frac{1}{n-1} \sum_{i=1}^n (\bar{v}_1^T \bar{x}_i)^2$$

$$= \frac{1}{n-1} \sum_{i=1}^n (\bar{v}_1^T \bar{x}_i) (\bar{x}_i^T \bar{v}_1)$$

$$= \frac{1}{n-1} \sum_{i=1}^n \bar{v}_1^T \bar{x}_i \bar{x}_i^T \bar{v}_1$$

$$= \bar{v}_1^T \underbrace{\left(\frac{1}{n-1} \sum_{i=1}^n \bar{x}_i \bar{x}_i^T \right)}_R \bar{v}_1$$

variance : $\bar{v}_1^T R \bar{v}_1$

$$\|\bar{v}_1\| = 1$$

$\Rightarrow \bar{v}_1 = \text{unit vector}$

in order to maximize variance :

max. $\bar{v}_1^T R \bar{v}_1$ — objective

subject to $\|\bar{v}_1\| = 1$.

$$\Rightarrow \|\bar{v}_1\|^2 = 1$$

$$\Rightarrow \bar{v}_1^T \bar{v}_1 = 1$$

↓
constraint

constrained
optimization
problem.

$$\max. \quad \bar{v}_1^T R \bar{v}_1$$

$$\text{s.t.} \quad \|\bar{v}_1\| = 1$$

constrained optimization
problem.

$$\bar{v}_1^T R \bar{v}_1 + \lambda (1 - \bar{v}_1^T \bar{v}_1)$$

→ Lagrange multiplier

$$\nabla f = \begin{bmatrix} \partial f / \partial \bar{v}_{11} \\ \partial f / \partial \bar{v}_{12} \\ \vdots \\ \partial f / \partial \bar{v}_{1m} \end{bmatrix}$$

↑
gradient
of Lagrangian

$$= \nabla (\bar{v}_1^T R \bar{v}_1 + \lambda (1 - \bar{v}_1^T \bar{v}_1))$$

$$\nabla f = 2R\bar{v}_1 - 2\lambda\bar{v}_1 = 0$$

↑
KKT condition for
extremum.

$$\Rightarrow R\bar{v}_1 = \lambda \bar{v}_1$$

↓
 $\bar{v}_1$ is eigen vector of R
corresponding to eigenvalue λ .

$$\bar{v}_1^T R \bar{v}_1 = \bar{v}_1^T \lambda \bar{v}_1$$

$$= \lambda \bar{v}_1^T \bar{v}_1$$

$$= \lambda \|\bar{v}_1\|^2 = \lambda \sim \text{maximize}$$

↑
To maximize

$\bar{v}_1$ = Eigenvector of R corresponding to largest eigenvalue.

Thus, the directions of the P principal components are $\bar{v}_1, \bar{v}_2, \dots, \bar{v}_p$ because R is PSD.
 (Note: $\bar{v}_1, \bar{v}_2, \dots, \bar{v}_p$ are orthogonal)
 eigenvectors of R corresponding to P largest eigenvalues.
 → projections are uncorrelated.

Principal components of $\bar{x}_i$ can be found as

$$\begin{bmatrix} \bar{v}_1^T \bar{x}_i \\ \bar{v}_2^T \bar{x}_i \\ \vdots \\ \bar{v}_p^T \bar{x}_i \end{bmatrix} = \begin{bmatrix} \bar{v}_1^T \\ \bar{v}_2^T \\ \vdots \\ \bar{v}_p^T \end{bmatrix} \bar{x}_i$$

$p < m$
 $p \times 1 \leftarrow \bar{x}_i = \bar{v}^T \bar{x}_i$

Dimensionality reduction

$$V = [\bar{v}_1 \ \bar{v}_2 \ \dots \ \bar{v}_p]$$

$m \times p$ matrix

Principal components of the data set
 are given as

$$\bar{v}^T \begin{bmatrix} \bar{x}_1 & \bar{x}_2 & \dots & \bar{x}_n \end{bmatrix} \quad m \times n$$

Feature
extraction

Dimensionality
reduction.

$$= \begin{bmatrix} \sqrt{\lambda_1} & \sqrt{\lambda_2} & \dots & \sqrt{\lambda_n} \end{bmatrix}_{p \times n}$$